COURSE CODE: CIC-307 • COMPUTER NETWORKS MINI-PROJECT

NetArchitect: Recommendation & Config Engine

Algorithmic Network Planning & Self-Explaining Architecture Engine

Subject Code: CIC-307 **Weight:** 100 Points **Submission Deadline:** 10 November 2026 **Format:** Team (2) or Individual

AI-Assisted Development & Academic Integrity Protocol
Generative AI (Claude, Copilot, ChatGPT, Gemini) is **explicitly allowed** to scaffold code, generate CLI parsers, or propose templates. However, students act as **Lead Network Architects and Validators**. You will be evaluated on your algorithmic boundary checks, edge-case failure detection, and technical defense against AI hallucinations.

1. Software Overview & Scenario Selector

Students will develop a software utility (CLI, web dashboard, or API) that ingests preset or custom enterprise network parameters, executes algorithmic rule engines, and outputs structured infrastructure recommendations alongside deployment-ready device configurations.

Profile 1: Campus LAN

Scale: 3 Buildings, <400m
Nodes: 1,600 endpoints, PoE APs
Focus: Cat6a vs MMF, 802.1Q VLANs, RSTP, Single-Area OSPF.

Profile 2: Metro MAN

Scale: HQ + 2 Remote (15–35km)
Nodes: 450 users, FinTech Colo
Focus: OS2 SMF, EVPL, Sub-5ms QoS, dual-homed failover.

Profile 3: Global WAN

Scale: 4 Cities + Cloud VPC
Nodes: 2,000 IoT / scanners
Focus: IPsec VPN, multi-area OSPF/BGP, route summarization.

Mandatory Feature: Explainable Decision Engine (Architectural Justifications)
The software must **dynamically explain why each choice was made** based on the supplied parameters. For example, if single-mode fiber is chosen, it must report: *"Selected OS2 Single-Mode Fiber because link distance (22 km) exceeds the 400m attenuation limit of OM4 Multimode Fiber."* Similarly, routing and subnetting choices must provide mathematical and RFC justifications.

2. System Architecture & Decision Flow



3. Algorithmic Decision Rules & Rationale Engine

The program must programmatically enforce constraints and output contextual justifications alongside choices:

Domain	Algorithmic Condition	Program Output & Justification Output
Physical Media (L1)	Run ≤ 100m (LAN access) Run 101–400m (Backbone) Run > 400m (MAN / WAN)	Recommend Cat6a (10GBASE-T) + Justification (802.3an limit). Recommend OM4 MMF (10GBASE-SR) + 850nm dispersion rationale. Recommend OS2 SMF (10GBASE-LR/ER) + optical budget rationale.
Boundary Error	Copper request > 100m	Raise `PhysicalDistanceViolationError` with signal attenuation warning.
VLSM Allocator	Input: Root prefix (e.g. `/16`) + host request array	Sort descending, allocate non-overlapping subnets, compute broadcast & gateways, output utilization efficiency rationale.
L3 Routing Protocol	Single autonomous system (<50 subnets) Multi-site / Multi-cloud boundary	Select OSPF Area 0 + justification (link-state fast convergence). Select Multi-Area OSPF or eBGP + routing domain isolation rationale.

4. Mandatory AI Defense & Code Auditing Log

Because generative AI accelerates code generation, students are required to deliver an **AI Audit Log** (`ai_audit_log.md`) documenting at least two concrete instances where AI suggested flawed networking code (e.g., bitwise mask errors, unconstrained switchport trunks, or incorrect optical budgets), detailing diagnosis and remediation.

5. Assessment Rubric (100 Points Total)

Evaluation Area	Grading Criteria	Points
1. Parameter Engine & Presets	Implements 3 preset scenarios and validates custom parameters (distance, density, resiliency). Gracefully handles boundary inputs and errors.	**15 Pts**
2. Explainable Decision Engine	Software outputs clear, natural-language engineering justifications for every selected media type, transceiver, routing protocol, and topology tier based on parameters.	**25 Pts**
3. Subnet Engine & Configs	Flawless dynamic bitwise VLSM allocation with zero overlap. Generates copy-paste ready Cisco IOS configurations for switches, routers, SVIs, and ACLs.	**25 Pts**
4. Automated Unit Testing	Test suite covering distance violations, address space exhaustion, and wildcard mask calculations across edge-case subnets.	**15 Pts**
5. AI Code Audit & Defense	Detailed documentation and technical post-mortem of two real AI misconfigurations or coding hallucinations, validated with packet flow analysis.	**20 Pts**

6. Submission Deliverables (Due: 10 November 2026)

- **Source Code Repository:** Clean codebase with CLI/Web interface, explainability engine, and unit test suite (`pytest` / `npm test`).
- **AI Code Audit Report:** `ai_audit_log.md` containing prompt, buggy AI code, technical analysis, and final fix.
- **Verification Exports:** Three exported configuration files demonstrating output from the 3 preset profiles loaded into Packet Tracer or GNS3.